Controlling Human Motion

Math 547 Final Project*

Deepan Islam and Krishna Saxena

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1 Introduction

1.1 A Heilmeir-style Abstract

Our goal for this study is to understand the minimal way to control human motion. Current models either operate in the noisy ambient space (e.g., Linear Time Invariant Control) or rely on simple models (i.e. PCA) which do not efficiently capture the interconnectedness of the human skeleto-muscular sytem. We use SINDy (Sparse Identification of Nonlinear Dynamics (Brunton, Proctor, and Kutz 2016)) to model the dynamics of human motion using higher-order polynomial and interaction terms. This enables us to discover that human locomotion (Lab 2003) can be controlled with $m^* = 1$ abstract actuator that acts according to the SINDy-identified dynamics.

1.2 Dataset description

We use the Locomotion subset of CMU's Motion Capture datset (Lab 2003). The dataset consists of $N = 15$ motion capture trials across three action classes — *jumping* (5 trials), *running* (5 trials), and *walking* (5 trials). Each trial consists of 38 joints' angular orientation, captured across 100 timesteps. Thus, each trial i can be considered as a matrix $\mathbf{X_i} \in \mathbb{R}^{100\times144}$. Our full study dataset $\mathbf{D}$ can be represented as a tensor $\mathbf{D} \in \mathbb{R}^{15\times100\times144}$.

2 Methods

In essence, we aim to identify the classic control framework for the MoCap dataset, using as few assumptions as possible:

$$\dot{x}(t) \;=\; f(x(t)) \;+\; \mathbf{B}\,u(t), \qquad x(t) \in \mathbb{R}^{144}, \quad u(t) \in \mathbb{R}^m, \tag{1}$$

where x is the joint-angle state vector, $f : \mathbb{R}^{144} \to \mathbb{R}^{144}$ captures the *autonomous dynamics* that drives all three types of natural actions, and $\mathbf{B}u$ is an *actuator control term* that drives the system between states.

2.1 Multi-Trajectory Dynamic Mode Decomposition (MTDMD)

Standard DMD identifies a linear operator A such that $x(t+1) \approx Ax(t)$ from a *single* trajectory. MTDMD (S. Anzaki and Y. Nakatsukasa 2024) extends this to N independent trajectories by solving the a least-squares optimization problem, to get $\hat{A} \in \mathbb{R}^{144\times144} : \forall i \in [N], x_i(t+1) \approx x_i(t)$.

*All data and code used in this analysis is available at https://github.com/atpabuser/Math547_FinalProject/.

Rank selection and latent basis. By taking the SVD of the pooled Hessian $H = \sum_\nu X_\nu X_\nu^\top \in \mathbb{R}^{n\times n} = U\Sigma V^\top$, we discover how much variance each spatial direction contributes to the dynamics across all trajectories. We retain the top-r left singular vectors as the *latent orthonormal basis*

$$U_r \in \mathbb{R}^{n\times r}, \qquad U_r^\top U_r = I_r,$$

choosing r so that the top-r singular values account for at least 97% of the total energy $\|H\|_F^2$.

We use this new basis to construct latent coordinates $z_\nu(t) = U_r^\top x_\nu(t) \in \mathbb{R}^r$ for each timeseries $x_\nu(t)$. We also project the full operator $\hat{A}$ onto this r-D basis $A_{\text{dmd}} = U_r^\top A^* U_r \in \mathbb{R}^{r\times r}$ to reveal a distilled dynamics:

$$z(t+1) = A_{\text{dmd}} z(t)$$

2.2 Sparse Identification of Nonlinear Dynamics (SINDy)

SINDy Brunton, Proctor, and Kutz 2016 identifies f in (1) by seeking a *sparse* representation in a fixed function library $\Theta : \mathbb{R}^r \to \mathbb{R}^p$.

Multi-trajectory pooling. For all 15 training trajectories we compute state-derivative pairs $\big(z_\mu(t),\, \dot{z}_\mu(t)\big)$ and stack them into

$$Z = \begin{bmatrix} z_1(0) \\ \vdots \\ z_{15}(98) \end{bmatrix} \in \mathbb{R}^{M\times r}, \quad \dot{Z} = \begin{bmatrix} \dot{z}_1(0) \\ \vdots \\ \dot{z}_{15}(98) \end{bmatrix} \in \mathbb{R}^{M\times r},$$

with $M = 15\times 99 = 1{,}485$ samples. Time derivatives are approximated by forward finite differences: $\dot{z}(t) \approx z(t+1) - z(t)$ (i.e. $\Delta t = 1$).

Library and regression. We use a degree-2 polynomial library in the latent $r = 3$-D space: $\Theta(z) = \big[1, z_1, z_2, z_3, z_1^2, z_1z_2, z_1z_3, z_2^2, z_2z_3, z_3^2\big] \in \mathbb{R}^p : p = 10$. This step reveals the importance of SVD in our pipeline: if we wanted to model nonlinear dynamics in the ambient space, there would be $\binom{144+d}{144}$ ($\approx 7{,}000$ for degree-2) terms, which is intractable for larger degrees.

We standardize each column of Z by its mean and standard deviation to get $\tilde{Z}$. SINDy then solves the sparse regression $\dot{\tilde{Z}} = \Theta(\tilde{Z})\Xi$ via Sequentially Thresholded Least Squares (STLSQ) with sparsity threshold $\lambda = 5\times 10^{-3}$. The result is a coefficient matrix $\Xi \in \mathbb{R}^{p\times r}$ with most entries set exactly to zero.

Autonomous dynamics model. The identified f in the latent coordinates is then fully identified as $f(z) = \Xi_0 \mathbf{1} + A_{\text{sindy}} z + o(\|z\|^2)$ where the linear block $A_{\text{sindy}} \in \mathbb{R}^{r\times r}$ is extracted from Ξ and un-scaled: $A_{\text{sindy}} = \operatorname{diag}(\sigma)\, \tilde{\Xi}_{\text{lin}} \operatorname{diag}(\sigma)^{-1}$.

2.3 Minimum-Actuator Controllability

Given a dynamics matrix $A \in \mathbb{R}^{r\times r}$ and an input matrix $B \in \mathbb{R}^{r\times m}$, the discrete-time system

$$z(t+1) = A\, z(t) + B\, u(t)$$

is *controllable* if and only if the Kalman controllability matrix

$$\mathcal{C}(A,B) = \big[B \quad AB \quad A^2B \quad \cdots \quad A^{r-1}B\big] \in \mathbb{R}^{r\times rm}$$

has rank r (Kalman 1960).

We restrict B to have columns drawn from the standard basis $\{\hat{e_1}, \hat{e_2}, \hat{e_3}\}$ of $\mathbb{R}^r$ ("latent-mode actuators") and seek the smallest m such that $\operatorname{rank}\mathcal{C}(A, B) = r$. This is solved by a greedy algorithm:

1. Initialise $B = []$, rank $= 0$.
2. For each candidate column e_j not yet selected, compute $\operatorname{rank}\mathcal{C}(A, [B\ e_j])$.
3. Add the e_j that gives the highest rank; update B.
4. Stop when rank $= r$.

Lifting to sensor space. The minimum input matrix in latent space $B_{\text{latent}} \in \mathbb{R}^{r\times m^*}$ lifts to the original 114-dimensional joint-angle space as

$$B \;=\; U_r\, B_{\text{latent}} \;\in\; \mathbb{R}^{n\times m^*}.$$

Each column of B is a unit vector describing the joint-angle actuation direction associated with that input channel.

3 Results

3.1 MTDMD: rank and latent basis

Figure 1: Scree and Energy Plots of Dynamics matrix $\hat{A}$

k	1	2	3
Cum. energy	90.2%	95.1%	97.4%

As we can see in 1, three modes suffice to explain 97% of the pooled Hessian energy, revealing that most of the data moves through a 3-dimensional manifold.

The reduced dynamics matrix is

$$A_{\text{dmd}} \;=\; \begin{pmatrix} 0.99975 & -0.00051 & 0.00203 \\ 0.00140 & 0.99702 & 0.00932 \\ -0.00031 & -0.00315 & 1.00216 \end{pmatrix},$$

with eigenvalues $\lambda_1 = 0.99944, \lambda_{2,3} = 0.99974 \pm 0.00491i$. Since all three eigenvalues lie just inside the unit circle, the trajectories neither grow nor decay significantly over 100 timesteps. The complex conjugate pair $\lambda_{2,3}$ corresponds to an oscillatory mode with angular period $\frac{2\pi}{\arctan(0.00491/0.99974)} \approx$ 1,280 timesteps, revealing that these features capture variance across samples (as opposed to within samples, where there are ≥ 1 repetitions of each action within the 100 timesteps).

3.2 Latent trajectories

Figure 2 shows all 15 latent trajectories $z_\mu(t) = U_r^\top x_\mu(t)$ in $\mathbb{R}^3$, coloured by class. The three action classes form three tight, well-separated clusters: jumping (red) occupies a region of large z_0, running (green) a region of large z_1, and walking (blue) occupies an intermediate region. Within each cluster, individual trial trajectories are nearly parallel short line segments, reflecting the fact that within a single trial the body's posture changes only slightly relative to the inter-class separation.

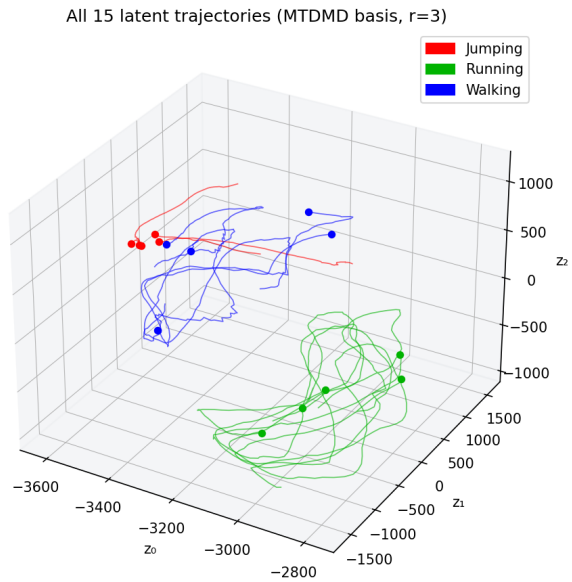

Figure 2: All 15 latent trajectories $z_\mu(t) = U_r^\top x_\mu(t)$ plotted in $\mathbb{R}^3$ and coloured by action class (red = jumping, green = running, blue = walking). Filled dots mark the initial condition $z_\mu(0)$.

3.3 SINDy: identified equations

With STLSQ threshold $\lambda = 5 \times 10^{-3}$ and a degree-2 polynomial library ($p = 10$ terms), SINDy identifies 14 non-zero terms out of 30 ($= r \times p$) possible entries, fitting 1,485 samples from all 15 trajectories simultaneously. The identified equations in scaled coordinates are:

$$\dot{\tilde{z}}_0 = -0.031 - 0.011\,\tilde{z}_0 + 0.009\,\tilde{z}_2 + 0.007\,\tilde{z}_0^2 + 0.005\,\tilde{z}_0\tilde{z}_2 + 0.016\,\tilde{z}_1^2 + 0.015\,\tilde{z}_2^2, \tag{2}$$

$$\dot{\tilde{z}}_1 = -0.009 + 0.006\,\tilde{z}_1 + 0.006\,\tilde{z}_2 + 0.009\,\tilde{z}_0^2 - 0.048\,\tilde{z}_0\tilde{z}_1 + 0.011\,\tilde{z}_0\tilde{z}_2 - 0.029\,\tilde{z}_1\tilde{z}_2, \tag{3}$$

$$\dot{\tilde{z}}_2 = 0. \tag{4}$$

Interestingly, the dynamics of all three actions' can be represented by two equations in the 3-D latent space!

The linear blocks, unscaled back to original latent coordinates, are:

$$A_{\text{sindy}} = \begin{pmatrix} -0.01060 & 0 & 0.00423 \\ 0 & 0.00615 & 0.00860 \\ 0 & 0 & 0 \end{pmatrix}, \qquad A_{\text{dmd}} - I = \begin{pmatrix} -0.000254 & -0.000513 & 0.002028 \\ 0.001404 & -0.002981 & 0.009317 \\ -0.000308 & -0.003151 & 0.002159 \end{pmatrix}. \tag{5}$$

3.4 Controllability and minimum actuators

The greedy algorithm 2.3 terminates after **one step** as adding $\hat{e_1}$ immediately achieves controllability rank 3. In fact, $\forall i \in [3], \mathcal{C}(A_{\text{dmd}}, \hat{e_i}) = 3 \Rightarrow \boxed{m^* = 1.}$

We thus choose, without loss of controllability, $\mathbf{B} = \hat{e_1}$ in latent space, or

$$\mathbf{B}_{\text{sensor}} = U_r \hat{e_1}$$

which is the first column of the MTDMD basis.

Figure 3 shows that B_{sensor} has large, roughly equal weights on features $\{14, 17, 20, 32, 35, 38\}$ where $|\mathbf{B}_{\text{sensor},j}| > 0.3$, meaning effective control of the full latent state requires coordinated actuation of fewer than 10 joint-angle channels.

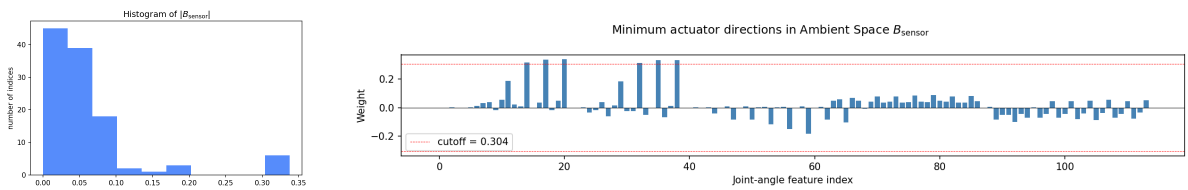

Figure 3: The minimum actuator $B_{\text{sensor}} = U_r[:, 0] \in \mathbb{R}^{114}$, shown as a bar chart over the 114 joint-angle feature indices. Each bar represents the weight of that feature in the optimal actuator direction.

4 Figure Descriptions

The result $m^* = 1$ means that — within the high-variance 3-dimensional MTDMD latent space — a *single scalar control signal* $u(t) \in \mathbb{R}$ suffices to transition the human body between any two latent states (e.g. from a walking configuration to a running configuration), provided the system is driven by the control input $Bu(t)$ along the direction $B_{\text{sensor}} \approx U_r[:, 0]$.

In practice, "controlling" real human motion means designing a trajectory $u(t)$ that moves $z(t)$ from the walking cluster to the running cluster in finite time. Since A_{dmd} is stable and the three clusters are separated by distances on the order of $\|z_{\text{run}} - z_{\text{walk}}\| \approx 800$ units, a finite-time minimum-energy control can be computed by solving a discrete-time LQR problem in the 3D latent space.

Interestingly, as seen in the reconstructed dynamics 4, the six most import joints for controlling the skeleton are in the left and right feet. This makes sense due to the nature of the MoCap data (jumping, running, and walking) and also due to the importance that locomotion plays in any physical activity. Based on this result, it would make sense to design prosthetic and robotic devices with accurate sensors and/or higher-grade materials on the ground.

Jumping Animation Running Animation Walking Animation

Figure 4: Reconstructed motion data with minimum actuators highlighted

References

Brunton, Steven L., Joshua L. Proctor, and J. Nathan Kutz (2016). "Discovering governing equations from data by sparse identification of nonlinear dynamical systems". In: *Proceedings of the National Academy of Sciences* 113.15, pp. 3932–3937. DOI: 10.1073/pnas.1517384113. URL: https://www.pnas.org/doi/abs/10.1073/pnas.1517384113.

Kalman, R.E. (1960). "On the general theory of control systems". In: *IFAC Proceedings Volumes* 1.1. 1st International IFAC Congress on Automatic and Remote Control, Moscow, USSR, 1960, pp. 491–502. ISSN: 1474-6670. DOI: https://doi.org/10.1016/S1474-6670(17)70094-8. URL: https://www.sciencedirect.com/science/article/pii/S1474667017700948.

Lab, CMU Motion Capture (2003). *The Motion Capture Database*. Carnegie Mellon University.

S. Anzaki, R. Yano and Y. Nakatsukasa (2024). "Multi-trajectory Dynamic Mode Decomposition". URL: https://jxiv.jst.go.jp/index.php/jxiv/preprint/download/602/2614.